ScAN - A Path to PIM Hardware-aware Network

Phase Roadmap

Phase 1 Float Training AllCNN-C CIFAR-100 FP32	**Phase 2** W4A4 QAT uniform W4 PACT activations integer simulator	**Phase 3** KD Finetune EfficientNet teacher SS / CS / TU	**Phase 4** PIM Row Sparsity 2/3 rows off ADC7 + digital BN current tuning
Phase 5 PCN Integration 3-iter fb/ff refine sensitive channels beta=1/8	**Phase 6** HW Noise Finetune real F/G/H LUT voltage readout noise model	Final Eval HW simulator full noise model multi-seed	

Phase Results

Phase	Experiment	Precision / path	Acc.	Notes
1	RGB AllCNN-C + BN	FP32 float	72.99 %	Standard CIFAR-100 RGB, 3x32x32, crop/flip augmentation; checkpoints/ALL_CNN_C_c100_rgb_aug_bn_refstyle.pth.
1	RAW/RGGB AllCNN-C	FP32 float	61.31 %	Current 4-channel RGGB BN baseline, raw_noise=True.
2	RAW/RGGB W4A4 SS	uniform W4A4 INT sim	56.61 %	Main noisy self-studying baseline; use uniform path going forward.
2	RAW/RGGB W4A4 SS + PACT	uniform I16/W4/A4 INT sim	60.01 %	Rerun with qkd_use_lutq=False; gain comes from PACT/I16 and simulator alignment, not non-uniform codebook.
3	QKD CS + PACT	I16/W4/A4 equal-spaced LUT-Q m=15	61.24 %	Co-studying result from simulator.
3	QKD TU + PACT	I16/W4/A4 equal-spaced LUT-Q m=15	61.66 %	Current best dense int4 simulator result.
4	PIM-QAT BN-only	spec W4/A4/ADC7	49.63 %	First completed PIM row-sparse run; target is 57-58, needs tuning.

Teacher / KD

Model / stage	Acc.	Notes
EfficientNetV2-L teacher, clean	77.30%	Verified on current H5 test split.
EfficientNetV2-L teacher, noisy	77.33%	Default noisy evaluation.
Co-studied teacher, noisy	78.70%	Earlier CS run improved teacher.
Uniform SS + PACT rerun	60.01%	I16/W4/A4, qkd_use_lutq=False, fixed integer simulator.
QKD CS rerun	61.24%	I16/W4/A4 equal-spaced LUT-Q + PACT, fixed simulator.
QKD TU rerun	61.66%	Best current dense int4 simulator result.

Current PIM Setting

Item	Current setting
Model	ALL-CNN-C, 4-channel RGGB input, BatchNorm; GBN and PCN disabled for the minimal pass.
Weights	Uniform 4-bit with deployed integer code range [-7,...,7] and precomputed per-output-channel scales.
Activations	First layer input is unsigned I16; hidden activations are unsigned A4 codes in [0,15] and values in [0,1].
Rows	Stage1.0 and logits are unmasked. Masked layers keep 48/144 rows; stage1.1 hybrid keeps 72/144 rows.
ADC	7-bit mid-rise tile ADC, fixed window W=K, no read noise in the current training run.
Simulator	PIM-QAT forward and PIM simulator are numerically matched on smoke tests; deployed inference uses integer weights/activations and precomputed constants.

Next

1. Improve Phase 4 PIM-QAT accuracy from 49.63% toward 57-58%.
2. Add pim_info/harware_model.tex voltage-domain readout as an optional hardware simulator once F/G/H and noise parameters are specified.
3. Add PCN only after the minimal BN row-sparse PIM path is stable.